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Method for manufacturing bearing ring member

Abstract

A method for manufacturing a bearing ring member includes an inversion step of sandwiching a workpiece including a workpiece body portion having an annular shape between a punch disposed on a first side in an axial direction and a die disposed on a second side opposite to the first side, and deforming the workpiece such that the workpiece body portion has a cylindrical shape. The inversion step includes, in the following order, a first step of deforming the workpiece using the punch and the die in a state where no sliding occurs between the workpiece and the die, and a second step of deforming the workpiece using the punch and the die in a state where a sliding occurs between the workpiece and the die.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a 35 U.S.C. § 371 national phase application of PCT/JP2022/042611, filed on Nov. 16, 2022, which claims priority to Japanese Patent Application No. 2021-189961, filed on Nov. 24, 2021 and Japanese Patent Application No. 2022-015064, filed on Feb. 2, 2022.

TECHNICAL FIELD

(2) One aspect of the present disclosure relates to a method for manufacturing a bearing ring member.

BACKGROUND ART

(3) Patent Literature 1 describes a method for manufacturing a ring member, which is used for manufacturing an inner ring or an outer ring of a bearing. In this manufacturing method, the ring member is formed through a step of performing inversion working in which a workpiece having an annular shape is sandwiched between a punch and a die and the direction of a cross section is changed by 90 degrees. Patent Literatures 2 to 5 also describe that a bearing ring member is formed by the same inversion working.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Unexamined Patent Publication No. 2006-341255 Patent Literature 2: Japanese Unexamined Patent Publication No. 2006-090407 Patent Literature 3: Japanese Unexamined Patent Publication No. 2006-097809 Patent Literature 4: Japanese Unexamined Patent Publication No. 2006-123003 Patent Literature 5: Japanese Unexamined Patent Publication No. 2020-022987

SUMMARY OF INVENTION

Technical Problem

(5) In the above-described manufacturing method, depending on the shape of the workpiece, it may be difficult to satisfactorily perform the inversion working. Examples of such a case include when the thickness of the workpiece is small, when the diameter or width of the workpiece is large or when the workpiece includes a flange portion.

(6) Therefore, an object of one aspect of the present disclosure is to provide a method for manufacturing a bearing ring member by which the inversion working of a workpiece can be satisfactorily performed.

Solution to Problem

(7) According to one aspect of the present disclosure, there is provided a method for manufacturing a bearing ring member, the method including an inversion step of sandwiching a workpiece including a workpiece body portion having an annular shape between a punch disposed on a first side in an axial direction and a die disposed on a second side opposite to the first side, and deforming the workpiece such that the workpiece body portion has a cylindrical shape. The inversion step includes, in the following order, a first step of deforming the workpiece using the punch and the die in a state where no sliding occurs between the workpiece and the die, and a second step of deforming the workpiece using the punch and the die in a state where a sliding occurs between the workpiece and the die.

(8) In the method for manufacturing a bearing ring member, the inversion step includes the first step of deforming the workpiece using the punch and the die in a state where no sliding occurs between the workpiece and the die, and the second step of deforming the workpiece using the punch and the die in a state where a sliding occurs between the workpiece and the die. Accordingly, in the first step, since no sliding occurs between the workpiece and the die, the workpiece can be satisfactorily stood up. In addition, in the second step, since the workpiece slides on the die, for example, the occurrence of unintended deformation such as the workpiece being bent can be suppressed. Therefore, according to the method for manufacturing a bearing ring member, the inversion working of the workpiece can be satisfactorily performed.

(9) A bending moment acting on the workpiece in the first step may be smaller than a predicted value M of a limiting bending moment of the workpiece body portion calculated by Formula (1). In this case, no sliding occurs between the workpiece and the die in the first step, and the workpiece can be satisfactorily stood up.

(10) [Formula1]

$$M = \frac{W_W \times t^2}{4} Y \quad (1)$$

(11) In Formula (1), W.sub.W is a width of the workpiece body portion, t is a thickness of the workpiece body portion, and Y is a yield stress of the workpiece body portion.

(12) In the second step, a second die different from a first die that is the die used in the first step may be used. In this case, satisfactory inversion working as described above can be realized by using different dies in the first step and the second step.

(13) The first die may include a first die R surface formed in a circular arc shape, and the second die may include a second die R surface formed in a circular arc shape, and an inclined surface formed on the second side of the second die R surface and inclined with respect to the axial direction. In the first step, the first die R surface may come into contact with the workpiece, and in the second step, after the second die R surface comes into contact with the workpiece, the inclined surface may come into contact with the workpiece. In this case, the workpiece can slide on the inclined surface in the second step, and satisfactory inversion working as described above can be realized.

(14) The first die may include a first die body portion formed in a columnar shape, and the second die may include a second die body portion formed in a columnar shape. In the first step, the workpiece may be sandwiched between the punch and the first die body portion, and in the second step, the workpiece may be sandwiched between the punch and the second die body portion. A diameter of the second die body portion may be smaller than a diameter of the first die body portion. In this case, the workpiece easily slides on the second die in the second step, so that satisfactory inversion working as described above can be realized.

(15) In the first step, the workpiece may be sandwiched between the punch and the first die in a state where a surface of the workpiece faces one of the first side and the second side, and in the second step, the workpiece may be sandwiched between the punch and the second die in a state where the surface of the workpiece faces the other of the first side and the second side. In this case as well, satisfactory inversion working as described above can be realized.

(16) A surface roughness of a second contact surface of the second die which comes into contact with the workpiece in the second step may be smaller than a surface roughness of a first contact surface of the first die which comes into contact with the workpiece in the first step. In this case, the workpiece easily slides on the second die in the second step, so that satisfactory inversion working as described above can be realized.

(17) A Rockwell C scale hardness of a second contact surface of the second die which comes into contact with the workpiece in the second step may be larger than a Rockwell C scale hardness of a first contact surface of the first die which comes into contact with the workpiece in the first step. In this case, the workpiece easily slides on the second die in the second step, so that satisfactory inversion working as described above can be realized.

(18) In the second step, the workpiece may be sandwiched between the punch and the die while the workpiece is pushed toward a side of the punch along a radial direction by a pushing member. In this case, the workpiece easily slides on the die in the second step, so that satisfactory inversion working as described above can be realized.

(19) In the second step, a hook punch including a hook portion may be used, and the workpiece may be sandwiched between the hook punch and the die by hooking the hook portion onto the workpiece. In this case, the workpiece easily slides on the die in the second step, so that satisfactory inversion working as described above can be realized.

(20) The workpiece may further include an inner flange portion extending from a radially inner edge of the workpiece body portion toward one side in the axial direction, and an outer flange portion extending from a radially outer edge of the workpiece body portion toward the one side in the axial direction. When the workpiece has such a shape, there is a concern that it is difficult to satisfactorily perform inversion working; however, according to the method for manufacturing a bearing ring member, even in such a case, the inversion working of the workpiece can be satisfactorily performed.
Advantageous Effects of Invention
(21) According to one aspect of the present disclosure, it is possible to provide the method for manufacturing a bearing ring member by which the inversion working of the workpiece can be satisfactorily performed.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIGS. **1**(*a*) and **1**(*b*) are cross-sectional views for describing a first step of an inversion step.
(2) FIGS. **2**(*a*) and **2**(*b*) are cross-sectional views for describing a second step of the inversion step.
(3) FIG. **3** is a graph showing an example of a bending moment acting on a workpiece in the inversion step.
(4) FIGS. **4**(*a*) to **4**(*d*) are cross-sectional views for describing a first step and a second step of a first modification example.
(5) FIGS. **5**(*a*) to **5**(*d*) are cross-sectional views for describing a first step and a second step of a second modification example.
(6) FIGS. **6**(*a*) to **6**(*d*) are cross-sectional views for describing a first step and a second step of a third modification example and a fourth modification example.
(7) FIGS. **7**(*a*) and **7**(*b*) are cross-sectional views for describing a second step of a fifth modification example.
(8) FIGS. **8**(*a*) and **8**(*b*) are cross-sectional views for describing a second step of a sixth modification example.
(9) FIGS. **9**(*a*) and **9**(*b*) are cross-sectional views for describing a second step of a seventh modification example.
DESCRIPTION OF EMBODIMENTS
(10) Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings. In the following description, the same reference signs are used for the same or corresponding elements, and duplicate descriptions will be omitted.
(11) In a method for manufacturing a bearing ring member according to the embodiment, as shown in FIGS. **1** and **2**, a bearing ring member **20** is manufactured from a workpiece **10**. FIGS. **1** and **2** show cross sections of the workpiece **10** and the ring member **20** which are parallel to an axial direction (a direction parallel to a central axis CL). In this example, the workpiece **10** and the ring member **20** have a substantially U-shaped cross-sectional shape.
(12) The ring member **20** is, for example, an inner ring member that can be used as an inner ring of a bearing. The manufactured ring member **20** itself may be used as an inner ring, or an inner ring may be manufactured by further processing the ring member **20**. The bearing to which the ring member **20** is applied may be any type of bearing, for example, a needle bearing, a cylindrical roller bearing, a tapered roller bearing, a ball bearing, or the like.
(13) The ring member **20** includes a main body portion **21** and a pair of flange portions **22** and **23**. The main body portion **21** is formed in a cylindrical shape, and has a cylindrical raceway surface **21***a* facing the outside in a radial direction. One flange portion **22** extends outward in the radial direction from an end portion on a first side S**1** (upper side in FIG. **2**) of the main body portion **21**

in the axial direction, and the other flange portion **23** extends outward in the radial direction from an end portion on a second side S**2** (side opposite to the first side S**1**) of the main body portion **21**. Each of the flange portions **22** and **23** is formed, for example, in a substantially annular plate shape.
(14) The workpiece **10** includes a workpiece body portion **11**, an inner flange portion **12**, and an outer flange portion **13**. The workpiece body portion **11** is formed in an annular plate shape, and has a surface **11***a* that becomes the raceway surface **21***a* of the ring member **20**. In this example, the surface **11***a* is an annular flat surface. The inner flange portion **12** extends from a radially inner edge of the workpiece body portion **11** toward one side (upper side in FIG. **1**) in the axial direction, and the outer flange portion **13** extends from a radially outer edge of the workpiece body portion **11** toward the one side in the axial direction. Namely, the inner flange portion **12** and the outer flange portion **13** protrude from the workpiece body portion **11** toward the same side. Each of the inner flange portion **12** and the outer flange portion **13** is formed, for example, in a substantially cylindrical shape.
(15) The method for manufacturing a bearing ring member of the embodiment includes an inversion step of sandwiching the workpiece **10** between a punch and a die, and deforming the workpiece **10** such that the workpiece body portion **11** has a cylindrical shape. The direction of a cross section of the workpiece **10** is changed by 90 degrees through the inversion step to form the workpiece body portion **11** having an annular plate shape into a cylindrical shape, so that the ring member **20** is formed. The inversion step includes a first step of deforming the workpiece **10** from an initial shape to a predetermined intermediate shape (FIG. **1**), and a second step of deforming the workpiece **10** from the intermediate shape to a final shape (FIG. **2**).
(16) As shown in FIG. **1**, in the first step, a first punch **30**A and a first die **40**A are used. The first punch **30**A and the first die **40**A have the common central axis CL, and each of the first punch **30**A and the first die **40**A has a uniform cross-sectional shape in the radial direction.
(17) The first punch **30**A includes a first punch body portion **31**A and a first punch protruding portion **32**A. The first punch body portion **31**A is formed in a substantially cylindrical shape having an axis parallel to the axial direction. A first punch R surface **35**A is formed at an inner edge portion on the second side S**2** of the first punch body portion **31**A. The first punch R surface **35**A is a curved surface formed by rounding a corner on the second side S**2** of the first punch body portion **31**A, and is formed in a circular are shape in a cross section parallel to the axial direction (FIG. **1**). The first punch protruding portion **32**A is formed on the first side S**1** of the first punch body portion **31**A, and protrudes inward from the first punch body portion **31**A in the radial direction. The first punch protruding portion **32**A is formed, for example, in a cylindrical shape.
(18) The first die **40**A includes a first die body portion **41**A and a first die protruding portion **42**A. The first die body portion **41**A is formed in a substantially columnar shape having an axis parallel to the axial direction. A first die R surface **45**A is formed at an outer edge portion on the first side S**1** of the first die body portion **41**A. The first die R surface **45**A is a curved surface formed by rounding a corner on the first side S**1** of the first die body portion **41**A, and is formed in a circular are shape in a cross section parallel to the axial direction (FIG. **1**). The first die protruding portion **42**A is formed on the second side S**2** of the first die body portion **41**A, and protrudes outward in the radial direction from the first die body portion **41**A. The first die protruding portion **42**A is formed, for example, in a cylindrical shape. Incidentally, in the present specification, the first punch body portion **31**A has a cylindrical shape, and the fact that the first die body portion **41**A has a columnar shape means that the first die body portion **41**A is formed in a substantially columnar shape to sandwich the workpiece **10** between the first die body portion **41**A and the first punch body portion **31**A from the inside in the radial direction, and includes a case where the first die body portion **41**A includes, for example, a hollow portion on the central axis CL as shown in FIG. **1**. This point also applies to a second die body portion **41**B.
(19) As shown in FIG. **2**, in the second step, a second punch **30**B and a second die **40**B are used. The second punch **30**B and the second die **40**B have the common central axis CL, and each of the

second punch **30**B and the second die **40**B has a uniform cross-sectional shape in the radial direction. In this example, the second punch **30**B is the same as the first punch **30**A, and the second die **40**B is different from the first die **40**A. The second punch **30**B includes a second punch body portion **31**B that is the same as the first punch body portion **31**A, and a second punch protruding portion **32**B that is the same as the first punch protruding portion **32**A. The second punch body portion **31**B has a second punch R surface **35**B corresponding to the first punch R surface **35**A.
(20) The second die **40**B includes the second die body portion **41**B and a second die protruding portion **42**B. The second die body portion **41**B is formed in a substantially columnar shape having an axis parallel to the axial direction. The second die body portion **41**B has the same configuration as the first die body portion **41**A, except that the second die body portion **41**B has an inclined surface **46**. Namely, the second die body portion **41**B has a second die R surface **45**B and the inclined surface **46**. The second die R surface **45**B is formed at an outer edge portion on the first side S**1** of the second die body portion **41**B. The second die R surface **45**B is a curved surface formed by rounding a corner on the first side S**1** of the second die body portion **41**B, and is formed in a circular arc shape in a cross section parallel to the axial direction (FIG. **2**).
(21) The inclined surface (tapered surface) **46** is formed on the second side S**2** of the second die R surface **45**B. The inclined surface **46** is inclined with respect to the axial direction to be further away from the central axis CL as the inclined surface **46** extends toward the second side S**2**. The second die protruding portion **42**B is formed on the second side S**2** of the second die body portion **41**B, and protrudes outward in the radial direction from the second die body portion **41**B. The second die protruding portion **42**B is formed, for example, in the same cylindrical shape as the first die protruding portion **42**A.
(22) In the inversion step, first, as shown in FIG. **1**, the workpiece **10** is sandwiched between the first punch **30**A and the first die **40**A, and the workpiece **10** is deformed to stand up toward the outside in the radial direction (first step). At the start of the first step, the first punch **30**A is disposed on the first side S**1** with respect to the first die **40**A, and the first die **40**A is disposed on the second side S**2** with respect to the first punch **30**A (FIG. **1**(*a*)). Then, by moving (descending) the first punch **30**A in the axial direction to bring the first punch **30**A closer to the first die **40**A, the workpiece **10** is sandwiched between the first punch **30** and the first die **40**A, and the workpiece **10** is deformed to stand up toward the outside in the radial direction (FIG. **1**(*b*)). In the first step, the first punch R surface **35**A and the first die R surface **45**A come into contact with the workpiece **10**. In the first step, the workpiece **10** is sandwiched between the first punch body portion **31**A and the first die body portion **41**A in a state where the surface **11***a* of the workpiece **10** faces the first side S**1**.
(23) In the first step, the workpiece **10** is deformed by the first punch **30**A and the first die **40**A in a state where no sliding occurs between the workpiece **10** and the first die **40**A (first die R surface **45**A). Accordingly, the workpiece **10** can be satisfactorily stood up. Whether or not sliding occurs can be determined using a predicted value M of a limiting bending moment of the workpiece **10** to be described below.
(24) When a bending moment (maximum bending moment) acting on the workpiece **10** in the first step is smaller than the predicted value M of the limiting bending moment of the workpiece **10** calculated by the following formula (2), no sliding occurs between the workpiece **10** and the first die **40**A, and the workpiece **10** stands up without sliding on the first die **40**A in the first step.
[Formula2]
(25)
$$M = \frac{W_W \times t^2}{4} Y \quad (2)$$
(26) In Formula (2), W.sub.W is a width of the workpiece body portion **11**, t is a thickness of the workpiece body portion **11**, and Y is a yield stress of the workpiece **10**. In this example, the width of the workpiece body portion **11** is a length between an inner edge of the inner flange portion **12** and an outer edge of the outer flange portion **13** in the radial direction. The yield stress is a

boundary stress that determines whether or not plastic deformation occurs in a material. When a stress larger than the yield stress acts, the material undergoes plastic deformation, and when a stress smaller than the yield stress acts, the material undergoes elastic deformation. Formula (2) is described, for example, in Hideaki Kudo's “Fundamental Mechanical Engineering Complete Book 3: Plasticity”, published by Morikita Publishing Co., Ltd., First edition, first printing on May 15, 1968, and First edition, 18th printing on Mar. 3, 1995. For example, a bending moment acting on the workpiece **10** can be theoretically calculated by calculating strain generated in the workpiece **10** from the amount of change in the width of the workpiece **10** (workpiece body portion **11**) in the inversion step, calculating stress from the strain, calculating energy from the stress, and calculating a bending moment from the energy.

(27) FIG. **3** is a graph showing an example of the bending moment acting on the workpiece **10** in the inversion step. The horizontal axis in FIG. **3** represents an inclination angle θ (°) of the workpiece **10** (workpiece body portion **11**) with respect to a horizontal direction (a direction perpendicular to the axial direction) (refer to FIG. **1**(*b*)), and the vertical axis represents the bending moment (N.Math.m) acting on the workpiece **10**. As shown in FIG. **3**, in this example, the bending moment is smaller than the predicted value M of the limiting bending moment within a range where the inclination angle θ is 0 degrees or more and approximately 40 degrees or less. In this case, no sliding occurs between the workpiece **10** and the first die **40**A within this range.

(28) In order to satisfactorily perform the inversion working of the workpiece **10**, the workpiece **10** needs to start sliding with respect to the die within a range where the bending moment acting on the workpiece **10** is smaller than the predicted value M of the limiting bending moment. In this example, the bending moment (maximum bending moment) acting on the workpiece **10** in the first step is smaller than the predicted value M of the limiting bending moment of the workpiece **10**. In other words, the shapes of the first punch **30**A and the first die **40**A, the descending stroke of the first punch **30**A, and the like are determined such that the bending moment acting on the workpiece **10** in the first step is smaller than the predicted value M. Accordingly, in the first step, no sliding occurs between the workpiece **10** and the first die **40**A, and the workpiece **10** can be satisfactorily stood up. As will be described below, in the second step, the workpiece **10** can slide on the second die **40**B, and the workpiece **10** can be satisfactorily deformed.

(29) In the second step, as shown in FIG. **2**, the workpiece **10** after processing in the first step is sandwiched between the second punch **30**B and the second die **40**B, and the workpiece **10** is deformed such that the workpiece body portion **11** has a cylindrical shape. In this example, the workpiece **10** is deformed such that the surface **11***a* of the workpiece body portion **11** faces the outside in the radial direction. In the second step, first, the workpiece **10** is sandwiched between the second punch R surface **35**B and the second die R surface **45**B (FIG. **2**(*a*)). Thereafter, the workpiece **10** is sandwiched between the second punch R surface **35**B and the inclined surface **46**. In such a manner, in the second step, after the second die R surface **45**B comes into contact with the workpiece **10**, the inclined surface **46** comes into contact with the workpiece **10**.

(30) Thereafter, when the second punch **30**B further descends, the second punch protruding portion **32**B comes into contact with the inner flange portion **12** of the workpiece **10**, and the workpiece **10** is pushed toward the second side S**2** by the second punch protruding portion **32**B. Thereafter, the workpiece **10** is disposed between the second punch body portion **31**B and the second die body portion **41**B (FIG. **2**(*b*)). Through the above-described steps, the workpiece body portion **11**, the inner flange portion **12**, and the outer flange portion **13** of the workpiece **10** become the main body portion **21**, the flange portion **22**, and the flange portion **23** of the ring member **20**, respectively, so that the ring member **20** is obtained.

(31) In the second step, the workpiece **10** is deformed by the second punch **30**B and the second die **40**B in a state where sliding occurs between the workpiece **10** and the second die **40**B (the second die R surface **45**B and the inclined surface **46**). Accordingly, for example, the occurrence of unintended deformation such as the workpiece **10** being bent can be suppressed. Namely, when no

sliding occurs between the workpiece **10** and the second die **40**B in the second step, there is a risk that the workpiece **10** may be bent into an unintended shape; however, since sliding occurs between the workpiece **10** and the second die **40**B in the second step, the occurrence of such deformation can be suppressed. Incidentally, in both the first step and the second step, no sliding occurs between the workpiece **10** and each of the first punch **30**A and the second punch **30**B.

(32) [Actions and Effects]

(33) In the method for manufacturing a bearing ring member of the embodiment, the inversion step includes the first step of deforming the workpiece **10** using the first punch **30**A and the first die **40**A in a state where no sliding occurs between the workpiece **10** and the first die **40**A, and the second step of deforming the workpiece **10** using the second punch **30**B and the second die **40**B in a state where sliding occurs between the workpiece **10** and the second die **40**B. Accordingly, in the first step, since no sliding occurs between the workpiece **10** and the first die **40**A, the workpiece **10** can be satisfactorily stood up. In addition, in the second step, since the workpiece **10** slides on the second die **40**B, for example, the occurrence of unintended deformation such as the workpiece **10** being bent can be suppressed. Therefore, according to the method for manufacturing a bearing ring member of the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed.

(34) The bending moment acting on the workpiece **10** in the first step is smaller than the predicted value M of the limiting bending moment of the workpiece body portion **11** calculated by Formula (2). Accordingly, no sliding occurs between the workpiece **10** and the first die **40**A in the first step, and the workpiece **10** can be satisfactorily stood up.

(35) In the second step, the second die **40**B different from the first die **40**A used in the first step is used. Accordingly, satisfactory inversion working as described above can be realized by using different dies in the first step and the second step.

(36) In the first step, the first die R surface **45**A comes into contact with the workpiece **10**, and in the second step, after the second die R surface **45**B comes into contact with the workpiece **10**, the inclined surface **46** comes into contact with the workpiece **10**. Accordingly, the workpiece **10** can slide on the inclined surface **46** in the second step, and satisfactory inversion working as described above can be realized.

(37) The workpiece **10** includes the inner flange portion **12** extending from the radially inner edge of the workpiece body portion **11** toward the one side in the axial direction, and the outer flange portion **13** extending from the radially outer edge of the workpiece body portion **11** toward the one side in the axial direction. When the workpiece **10** has such a shape, there is a risk that it is difficult to satisfactorily perform inversion working; however, according to the method for manufacturing a bearing ring member of the embodiment, even in such a case, the inversion working of the workpiece **10** can be satisfactorily performed.

(38) [Modification Examples]

(39) FIG. **4** is a cross-sectional view for describing a first step and a second step of a first modification example. The first step of the first modification example is the same as the first step of the embodiment. In the first modification example, the second die body portion **41**B of the second die **40**B does not include the inclined surface **46**. A diameter D**2** of the second die body portion **41**B is smaller than a diameter D**1** of the first die body portion **41**A.

(40) In the first modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. In addition, since the diameter D**2** of the second die body portion **41**B is smaller than the diameter D**1** of the first die body portion **41**A, the workpiece **10** easily slides on the second die **40**B in the second step, so that satisfactory inversion working can be realized.

(41) FIG. **5** is a cross-sectional view for describing a first step and a second step of a second modification example. In the second modification example, contrary to the embodiment, the first punch body portion **31**A of the first punch **30**A is formed in a substantially columnar shape, and the

first die body portion **41**A of the first die **40**A is formed in a substantially cylindrical shape. Similarly to the embodiment, the second punch body portion **31**B of the second punch **30**B is formed in a substantially cylindrical shape, and the second die body portion **41**B of the second die **40**B is formed in a substantially columnar shape. Similarly to the first modification example, the second die body portion **41**B does not include the inclined surface **46**.
(42) In the first step of the second modification example, the workpiece **10** is sandwiched between the first punch body portion **31**A and the first die body portion **41**A in a state where the surface **11***a* of the workpiece **10** faces the second side S**2**. In the second step, the workpiece **10** is sandwiched between the second punch body portion **31**B and the second die body portion **41**B in a state where the surface **11***a* of the workpiece **10** faces the first side S**1**. Namely, in the second modification example, a step of inverting the direction of the workpiece **10** in the axial direction is included between the first step and the second step, and the direction of the workpiece **10** in the axial direction is opposite between the first step and the second step.
(43) In the second modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. Incidentally, in the second modification example, the first punch body portion **31**A and the second die body portion **41**B may be formed in a substantially cylindrical shape, and the first die body portion **41**A and the second punch body portion **31**B may be formed in a substantially columnar shape. Alternatively, the first punch body portion **31**A and the second punch body portion **31**B may be formed in a substantially cylindrical shape, and the first die body portion **41**A and the second die body portion **41**B may be formed in a substantially columnar shape. On the contrary, the first punch body portion **31**A and the second punch body portion **31**B may be formed in a substantially columnar shape, and the first die body portion **41**A and the second die body portion **41**B may be formed in a substantially cylindrical shape. In addition, the direction of the workpiece **10** in the axial direction may be opposite between the first step and the second step; in the first step, the workpiece **10** may be sandwiched between the first punch body portion **31**A and the first die body portion **41**A in a state where the surface **11***a* of the workpiece **10** faces the first side S**1**, and in the second step, the workpiece **10** may be sandwiched between the second punch body portion **31**B and the second die body portion **41**B in a state where the surface **11***a* of the workpiece **10** faces the second side S**2**.
(44) FIG. **6** is a cross-sectional view for describing a first step and a second step of a third modification example and a fourth modification example. In the third modification example, a surface roughness of a second contact surface **47**B of the second die **40**B which comes into contact with the workpiece **10** in the second step is smaller than a surface roughness of a first contact surface **47**A of the first die **40**A which comes into contact with the workpiece **10** in the first step. The surface roughness is an arithmetic average roughness Ra (JIS B 0601 4.2.1). Similarly to the first modification example, the second die body portion **41**B of the third modification example does not include the inclined surface **46**. In this example, the first contact surface **47**A is the first die R surface **45**A, and the second contact surface **47**B is the second die R surface **45**B. The surface roughness of the first contact surface **47**A can be adjusted, for example, by forming a mill scale film or performing lathing or grinding (forming lathe marks or grinding marks). The surface roughness of the second contact surface **47**B can be adjusted, for example, by performing grinding or lapping or forming a coating layer. By selectively performing these processes, the surface roughness of the first contact surface **47**A and the second contact surface **47**B can be adjusted such that the surface roughness of the second contact surface **47**B is smaller than the surface roughness of the first contact surface **47**A.
(45) In the third modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. In addition, since the surface roughness of the second contact surface **47**B is smaller than the surface roughness of the first contact surface **47**A, the workpiece **10** easily slides on the second die **40**B in the second step, so that satisfactory inversion working can be realized.

(46) Subsequently, the fourth modification example will be described with reference to FIG. **6**. In the fourth modification example, instead of adjusting the surface roughness of the first contact surface **47**A and the second contact surface **47**B, the Rockwell C scale hardness is adjusted. Specifically, a Rockwell C scale hardness of the second contact surface **47**B is larger than a Rockwell C scale hardness of the first contact surface **47**A. The fourth modification example is the same as the third modification example except for this point. In this example, by making the first die R surface **45**A from a material prone to wear, whereas making the second die R surface **45**B from a material resistant to wear, the Rockwell C scale hardness of the second contact surface **47**B is made larger than the Rockwell C scale hardness of the first contact surface **47**A. The material of the first contact surface **47**A may be, for example, steel, die steel (for example, SKD11), or the like. The material of the second contact surface **47**B may be, for example, cemented carbide, high-speed steel (for example, SKH51), or the like. The Rockwell C scale hardness of the second contact surface **47**B may be adjusted by forming a coating layer on the second contact surface **47**B. The Rockwell C scale hardness is a hardness defined by JIS Z 2245.
(47) In the fourth modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. In addition, since the Rockwell C scale hardness of the second contact surface **47**B is larger than the Rockwell C scale hardness of the first contact surface **47**A, the workpiece **10** easily slides on the second die **40**B in the second step, so that satisfactory inversion working can be realized.
(48) FIG. **7** is a cross-sectional view for describing a second step of a fifth modification example. For example, a first step of the fifth modification example is the same as the first step of the embodiment. In the second step of the fifth modification example, the workpiece **10** is sandwiched between the second punch **30**B and the second die **40**B while the workpiece **10** is pushed from the inside toward the outside in the radial direction (while the workpiece **10** is pushed toward a second punch **30**B side along the radial direction) by a pushing member **51**. The pushing member **51** includes, for example, a plurality of (for example, eight) block members **51***a* arranged along a circumferential direction. Each block member **51***a* is disposed on the second die **40**B, and is movable in the radial direction along the second die **40**B. As the block members **51***a* moves outward in the radial direction, the pushing member **51** pushes the workpiece **10** outward in the radial direction. As the pushing force, for example, a force from any mechanism or power source such as a spring, hydraulic pressure, air, a motor, or a cam may be used. Similarly to the first modification example, the second die body portion **41**B of the fifth modification example does not include the inclined surface **46**.
(49) In the fifth modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. In addition, since the workpiece **10** is sandwiched between the second punch **30**B and the second die **40**B while the workpiece **10** is pushed toward the second punch **30**B along the radial direction by the pushing member **51**, the workpiece **10** easily slides on the second die **40**B in the second step, so that satisfactory inversion working can be realized. Incidentally, in the fifth modification example, the first punch body portion **31**A and the second punch body portion **31**B may be formed in a substantially columnar shape, and the first die body portion **41**A and the second die body portion **41**B may be formed in a substantially cylindrical shape. In this case, in the second step, the workpiece **10** may be sandwiched between the second punch **30**B and the second die **40**B while the workpiece **10** is pushed from the outside toward the inside in the radial direction (while the workpiece **10** is pushed toward the second punch **30**B side along the radial direction) by the pushing member **51**.
(50) FIG. **8** is a cross-sectional view for describing a second step of a sixth modification example. For example, a first step of the sixth modification example is the same as the first step of the embodiment. In the second step of the sixth modification example, the second punch **30**B (hook punch) including a hook portion **52** is used, and the workpiece **10** is sandwiched between the second punch **30**B and the second die **40**B by hooking the hook portion **52** onto the workpiece **10**.

The second punch **30**B includes a plurality of (for example, eight) divided portions **53** arranged along the circumferential direction. Each divided portion **53** is movable in the radial direction.
(51) In this example, the hook portion **52** is provided in each divided portion **53**. Namely, each divided portion **53** includes a main body portion **53***a* and a protruding portion **53***b* protruding inward in the radial direction from an end portion on the first side S**1** of the main body portion **53***a*, and the protruding portion **53***b* functions as the hook portion **52**. In other words, the hook portion **52** includes a plurality of portions (protruding portions **53***b*) arranged along the circumferential direction.
(52) In the second step, the hook portion **52** moves toward the lower side (second side S**2**) while moving inward in the radial direction. Specifically, a cam **54** is disposed below the second punch **30**B. The cam **54** has a guide surface **54***a* inclined to face the second punch **30**B. The second punch **30**B (each divided portion **53**) has an inclined surface **30**Ba corresponding to the guide surface **54***a*. The inclined surface **30**Ba and the guide surface **54***a* are inclined to approach the central axis CL as the inclined surface **30**Ba and the guide surface **54***a* extend downward. During inversion working, the second punch **30**B is pushed downward by a pushing jig J, the inclined surface **30**Ba comes into contact with the guide surface **54***a*, and the inclined surface **30**Ba is guided by the guide surface **54***a*. Accordingly, the second punch **30**B moves diagonally inward in the radial direction and downward.
(53) In the second step, due to such movement of the second punch **30**B, as shown in FIG. **8**(*a*), the hook portion **52** is hooked onto the outer flange portion **13**, and the outer flange portion **13** is drawn inward in the radial direction. In the hooking operation, the hook portion **52** comes into contact with an inner surface of the outer flange portion **13**. In parallel with the hooking operation, similarly to the fifth modification example, the inner flange portion **12** is pushed outward by the pushing member **51** on a radially inner side. Thereafter, as shown in FIG. **8**(*b*), the second punch **30**B moves straight downward, thereby completing the inversion working.
(54) In the sixth modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. In addition, since the workpiece **10** is sandwiched between the second punch **30**B and the second die **40**B by hooking the hook portion **52** onto the workpiece **10**, the workpiece **10** easily slides on the second die **40**B in the second step, so that satisfactory inversion working can be realized.
(55) FIG. **9** is a cross-sectional view for describing a second step of a seventh modification example. The seventh modification example is the same as the sixth modification example except for points to be described below. In the seventh modification example, a contact surface **51***b* of the pushing member **51** with the workpiece **10** is inclined with respect to the axial direction to be further away from the central axis CL as the contact surface **51***b* extends toward the lower side (second side S**2**). In addition, in the seventh modification example, the second punch **30**B includes an annular member **55**; the hook portion **52** that is movable with respect to the annular member **55**; and a guide member **56** that guides the movement of the hook portion **52**. The annular member **55** is formed of, for example, one member formed in an annular shape in a plan view.
(56) The hook portion **52** has an inclined surface **52***a*, and the guide member **56** has an inclined guide surface **56***a*. The inclined surface **52***a* and the guide surface **56***a* are inclined to approach the central axis CL as the inclined surface **52***a* and the guide surface **56***a* extend downward. During inversion working, as shown in FIG. **9**(*a*), the hook portion **52** is pushed downward by the pushing jig J, the inclined surface **52***a* is guided by the guide surface **56***a*, and the hook portion **52** moves diagonally inward in the radial direction and downward. Accordingly, the hook portion **52** protrudes from the annular member **55**. In parallel with the protruding operation, the inner flange portion **12** is pushed outward by the pushing member **51** on the radially inner side. Thereafter, as shown in FIG. **9**(*b*), a tip portion **52***b* of the hook portion **52** is hooked onto the outer flange portion **13**, and pushes the inner surface of the outer flange portion **13** downward. Accordingly, the inversion working is completed.

(57) In the seventh modification example as well, similarly to the embodiment, the inversion working of the workpiece **10** can be satisfactorily performed. In addition, since the workpiece **10** is sandwiched between the second punch **30**B and the second die **40**B by hooking the hook portion **52** onto the workpiece **10**, the workpiece **10** easily slides on the second die **40**B in the second step, so that satisfactory inversion working can be realized. Incidentally, in the sixth modification example and the seventh modification example, the pushing member **51** may be omitted.
(58) The present disclosure is not limited to the embodiment and the modification examples. For example, the material and shape of each configuration are not limited to the material and shape described above, and various materials and shapes can be adopted. The workpiece **10** may have any shape, and may include, for example, only the workpiece body portion **11** without including the inner flange portion **12** and the outer flange portion **13**.
(59) The ring member **20** may be, for example, an outer ring member that can be used as an outer ring of a bearing. In this case, the manufactured ring member **20** itself may be used as an outer ring, or an outer ring may be manufactured by further processing the ring member **20**. When the ring member **20** is an outer ring member, the main body portion **21** has the raceway surface **21***a* facing the inside in the radial direction, and the pair of flange portions **22** and **23** extend inward in the radial direction from the main body portion **21**. In the first step, the workpiece **10** is deformed to stand up toward the inside in the radial direction, and in the second step, the workpiece **10** is deformed such that the surface **11***a* of the workpiece body portion **11** faces the inside in the radial direction. In this case as well, the workpiece **10** is deformed by the inversion step such that the workpiece body portion **11** has a cylindrical shape. In this case, the first punch body portion **31**A and the second punch body portion **31**B may be formed in a substantially columnar shape, and the first die body portion **41**A and the second die body portion **41**B may be formed in a substantially cylindrical shape.
(60) In the embodiment, the workpiece **10** is sandwiched between the punch and the die by bringing the punch closer to the die; however, the punch and the die may be moved relative to each other, for example, the workpiece **10** may be sandwiched between the punch and the die by moving the punch and the die to approach each other.
REFERENCE SIGNS LIST
(61) **10**: workpiece, **11**: workpiece body portion, **11***a*: surface, **12**: inner flange portion, **13**: outer flange portion, **20**: ring member (bearing ring member), **30**A: first punch, **30**B: second punch, **40**A: first die, **41**A: first die body portion, **40**B: second die, **41**B: second die body portion, **45**A: first die R surface, **45**B: second die R surface, **46**: inclined surface, **47**A: first contact surface, **47**B: second contact surface, **51**: pushing member, **52**: hook portion, S**1**: first side, S**2**: second side.

Claims

1. A method for manufacturing a bearing ring member, the method comprising: an inversion step of sandwiching a workpiece including a workpiece body portion having an annular shape between one or more punches disposed on a first side in an axial direction and a plurality of dies disposed on a second side opposite to the first side, and deforming the workpiece using the one or more punches and the plurality of dies such that the workpiece body portion has a cylindrical shape, wherein the inversion step includes, in the following order, a first step of deforming the workpiece using a first die of the plurality of dies in a state where no sliding occurs between the workpiece and the first die, and a second step of deforming the workpiece using a second die of the plurality of dies in a state where a sliding occurs between the workpiece and the second die, and a bending moment acting on the workpiece in the first step is smaller than a predicted value M of a limiting bending moment of the workpiece body portion calculated by Formula (1), $M = \frac{W_W \times t^2}{4} Y \quad (1)$ where W.sub.W is a width of the workpiece body portion, t is a thickness of the workpiece body portion,

and Y is a yield stress of the workpiece body portion.

2. The method for manufacturing a bearing ring member according to claim 1, wherein the first die includes a first die R surface formed in a circular arc shape, the second die includes a second die R surface formed in a circular arc shape, and an inclined surface formed on the second side of the second die R surface and inclined with respect to the axial direction, in the first step, the first die R surface comes into contact with the workpiece, and in the second step, after the second die R surface comes into contact with the workpiece, the inclined surface comes into contact with the workpiece.

3. The method for manufacturing a bearing ring member according to claim 1, wherein the first die includes a first die body portion formed in a columnar shape, the second die includes a second die body portion formed in a columnar shape, in the first step, the workpiece is sandwiched between one punch of the one or more punches and the first die body portion, in the second step, the workpiece is sandwiched between the one punch and the second die body portion, and a diameter of the second die body portion is smaller than a diameter of the first die body portion.

4. The method for manufacturing a bearing ring member according to claim 1, wherein in the first step, the workpiece is sandwiched between one punch of the one or more punches and the first die in a state where a surface of the workpiece faces one of the first side and the second side, and in the second step, the workpiece is sandwiched between the one punch and the second die in a state where the surface of the workpiece faces the other of the first side and the second side.

5. The method for manufacturing a bearing ring member according to claim 1, wherein a surface roughness of a second contact surface of the second die which comes into contact with the workpiece in the second step is smaller than a surface roughness of a first contact surface of the first die which comes into contact with the workpiece in the first step.

6. The method for manufacturing a bearing ring member according to claim 1, wherein a Rockwell C scale hardness of a second contact surface of the second die which comes into contact with the workpiece in the second step is larger than a Rockwell C scale hardness of a first contact surface of the first die which comes into contact with the workpiece in the first step.

7. The method for manufacturing a bearing ring member according to claim 1, wherein in the second step, while the workpiece is sandwiched in the axial direction, the workpiece is pushed along a radial direction by a pushing member.

8. The method for manufacturing a bearing ring member according to claim 1, wherein in the second step, a hook punch including a hook portion is used, and the workpiece is sandwiched between the hook punch and the second die by hooking the hook portion onto the workpiece.

9. The method for manufacturing a bearing ring member according to claim 1, wherein the workpiece further includes an inner flange portion extending from a radially inner edge of the workpiece body portion toward one side in the axial direction, and an outer flange portion extending from a radially outer edge of the workpiece body portion toward the one side in the axial direction.

10. The method for manufacturing a bearing ring member according to claim 1, wherein the first die includes a first die body portion formed in a columnar shape, the second die includes a second die body portion formed in a columnar shape, in the first step, the workpiece is sandwiched between a first punch of the one or more punches and the first die body portion, in the second step, the workpiece is sandwiched between a second punch of the one or more punches and the second die body portion, and a diameter of the second die body portion is smaller than a diameter of the first die body portion.

11. The method for manufacturing a bearing ring member according to claim 1, wherein in the first step, the workpiece is sandwiched between a first punch of the one or more punches and the first die in a state where a surface of the workpiece faces one of the first side and the second side, and in the second step, the workpiece is sandwiched between a second punch of the one or more punches and the second die in a state where the surface of the workpiece faces the other of the first side and

the second side.

12. A method for manufacturing a bearing ring member using an inversion process, the method comprising: sandwiching a workpiece including a workpiece body portion having an annular shape between one or more punches disposed on a first side in an axial direction and a plurality of dies disposed on a second side opposite to the first side; deforming the workpiece a first time, while sandwiching the workpiece, using a first die of the plurality of dies in a state where no sliding occurs between the workpiece and the first die; and deforming the workpiece a second time, while sandwiching the workpiece, using a second die of the plurality of dies in a state where a sliding occurs between the workpiece and the second die, wherein a surface roughness of a second contact surface of the second die which comes into contact with the workpiece is smaller than a surface roughness of a first contact surface of the first die which comes into contact with the workpiece.

13. The method for manufacturing a bearing ring member according to claim 12, wherein the workpiece body portion has a cylindrical shape after the workpiece is deformed the second time.

14. The method for manufacturing a bearing ring member according to claim 12, wherein during the first time that the workpiece is deformed, the workpiece is sandwiched between one punch of the one or more punches and the first die, and during the second time that the workpiece is deformed, the workpiece is sandwiched between the one punch and the second die.

15. The method for manufacturing a bearing ring member according to claim 12, wherein during the first time that the workpiece is deformed, the workpiece is sandwiched between a first punch of the one or more punches and the first die, and during the second time that the workpiece is deformed, the workpiece is sandwiched between a second punch of the one or more punches and the second die.

16. The method for manufacturing a bearing ring member according to claim 12, wherein the second time the workpiece is deformed, a hook punch including a hook portion is used, and the workpiece is sandwiched between the hook punch and the second die by hooking the hook portion onto the workpiece.

17. The method for manufacturing a bearing ring member according to claim 12, wherein the workpiece further includes an inner flange portion extending from a radially inner edge of the workpiece body portion toward one side in the axial direction, and an outer flange portion extending from a radially outer edge of the workpiece body portion toward the one side in the axial direction.

18. A method for manufacturing a bearing ring member using an inversion process, the method comprising: sandwiching a workpiece including a workpiece body portion having an annular shape between one or more punches disposed on a first side in an axial direction and a plurality of dies disposed on a second side opposite to the first side; deforming the workpiece a first time, while sandwiching the workpiece, using a first die of the plurality of dies in a state where no sliding occurs between the workpiece and the first die; and deforming the workpiece a second time, while sandwiching the workpiece, using a second die of the plurality of dies in a state where a sliding occurs between the workpiece and the second die, wherein a Rockwell C scale hardness of a second contact surface of the second die which comes into contact with the workpiece is larger than a Rockwell C scale hardness of a first contact surface of the first die which comes into contact with the workpiece.

19. The method for manufacturing a bearing ring member according to claim 18, wherein during the first time that the workpiece is deformed, the workpiece is sandwiched between one punch of the one or more punches and the first die, during the second time that the workpiece is deformed, the workpiece is sandwiched between the one punch and the second die, and the workpiece body portion has a cylindrical shape after the workpiece is deformed the second time.

20. The method for manufacturing a bearing ring member according to claim 18, wherein during the first time that the workpiece is deformed, the workpiece is sandwiched between a first punch of the one or more punches and the first die, during the second time that the workpiece is deformed,

the workpiece is sandwiched between a second punch of the one or more punches and the second die, and the workpiece body portion has a cylindrical shape after the workpiece is deformed the second time.